

Dot Net Deserialization

Dot Net Deserialization - Author not stated, Metasploit Documentation Penetration Testing Software, Pen Testing Security.

- Published: date not stated
- Original: <https://docs.metasploit.com/docs/development/developing-modules/libraries/deserialization/dot-net-deserialization.html>
- Preserved from: <https://docs.metasploit.com/docs/development/developing-modules/libraries/deserialization/dot-net-deserialization.html> (preserved-copy) on 2026-08-04
- Licence: unknown

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Content

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Metasploit includes a library for leveraging .NET deserialization attacks. Using it within a module is very straight forward, the module author just needs to know two things: the gadget chain and the formatter. The library uses the same names for each of these values as the [YSoSerial.NET](#) project for compatibility, although the Metasploit library only supports a subset of the functionality.

[]() Support Matrix

The following table outlines the supported gadget chains, formatters and the compatibility of each.

```
| Gadget Chain Name BinaryFormatter LosFormatter SoapFormatter |  
ClaimsPrincipal YesYesYes | TextFormatting RunProperties YesYesYes |  
TypeConfuseDelegate YesYesNo | WindowsIdentity YesYesYes |||||
```

[]() Basic Usage

The library is located in `Msf::Util::DotNetDeserialization` and contains the following methods which are intended for use by module authors.

—

```
#generate(cmd, gadget chain:, formatter:)
```

This function will generate a serialized payload to execute the specified operating system command *cmd*. The command is serialized using the specified *gadget_chain* and formatted with the specified *formatter*. The *gadget_chain* and *formatter* options will be specific to the vulnerability that is being executed. This functions returns a string.

-

```
#generate_formatted(stream, formatter:)
```

Format a `SerializedStream` object, as created by `#generate_gadget_chain`. The *stream* will be formatted using the specified *formatter* and returned as a string.

-

```
#generate_gadget_chain(cmd, gadget_chain:)
```

Create a gadget chain to run the specified operating system command *cmd*. This returns a `SerializedStream` object which can be inspected and modified but must formatted (using `#generate_formatted`) before it is useful.

`#generate` is the primary function and is functionally equivalent to the following. In the future the `#generate_*` functions may contain additional options specific to their respective chain or formatter.

```
stream = generate_gadget_chain(cmd, gadget_chain)
formatted = generate_formatted(stream, formatter)
```

[]() Example Usage

The following example uses the `TextFormattingRunProperties` gadget chain formatted with the `LosFormatter`.

```
serialized = ::Msfp::Util::DotNetDeserialization.generate(
  cmd, # this is the Operating System command to run
  gadget_chain: :TextFormattingRunProperties,
  formatter: :LosFormatter
)
```

[]() Command Line Tool

The library also has an interface available as a standalone command line tool which is suitable for creating payloads for single-use research purposes. This tool `dot_net.rb` is available in the `tools/payloads/ysoserial` directory. The arguments for this tool are aligned with those of [YSoSerial.NET](#), allowing the arguments of basic invocations to be the same. It should be noted however that the `[supported]()` gadgets and formatters are not the same.

Help output:

Usage: `./dot_net.rb [options]`

Generate a .NET deserialization payload that will execute an operating system command **using** the specified gadget chain and formatter.

Available formatters:

- * BinaryFormatter
- * LosFormatter
- * SoapFormatter

Available gadget chains:

- * ClaimsPrincipal
- * DataSet
- * DataSetTypeSpooF
- * ObjectDataProvider
- * TextFormattingRunProperties
- * TypeConfuseDelegate
- * WindowsIdentity

Available HMAC algorithms: SHA1, HMACSHA256, HMACSHA384, HMACSHA512, MD5

Examples:

```
./dot_net.rb -c "net user msf msf /ADD" -f BinaryFormatter -g TypeConfuseDelegate -o base64
```

```
./dot_net.rb -c "calc.exe" -f LosFormatter -g TextFormattingRunProperties \  
--viewstate-validation-key deadbeef --viewstate-validation-algorithm SHA1
```

General options:

<code>-h, --help</code>	Show this message
<code>-c, --command <String></code>	The command to run
<code>-f, --formatter <String></code>	The formatter to use (default : BinaryFormatter)
<code>-g, --gadget <String></code>	The gadget chain to use (default : TextFormattingRunProperties)
<code>-o, --output <String></code>	The output format to use (default : raw, see: --list- output-formats)
<code>--list-output-formats</code>	List available output formats, for use with --output

ViewState related options:

- `--viewstate-generator <String>`
The ViewState generator **string** to **use**
- `--viewstate-validation-algorithm <String>`
The validation algorithm (**default**: SHA1, see:
Available HMAC algorithms)

```
--viewstate-validation-key
```

```
<HexString>
```

The validationKey **from** the web.config file

The `-g / --gadget` option maps to the `gadget_chain` argument for the generate functions while the `-f / --formatter` arguments maps to the `formatter` argument.

[]() Making Changes

Adding new gadget chains and formatters involves creating a new file in the respective library directory: `lib/msf/util/dot_net_deserialization`. The “native” gadget chain type is implemented following the `MS-NRBF` format and the `Bindata` records as defined in `types/` subdirectory. Once the new gadget chain or formatter is implemented, it needs to be added to the main library file (`dot_net_deserialization.rb`).

Since serialization chain generate is deterministic, a `unit test` should be added for any new gadget chain to ensure that the checksum of the `BinaryFormatter` representation is consistent.

[]() Further Reading

Since the .NET deserialization gadgets run operating system commands, the following resources can be helpful for module developers to deliver native payloads such as Meterpreter.

- [How to use command stagers](#)
- [How to use Powershell in an exploit](#)

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